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Addressing Shoulder Weakness in Individuals With Rotator Cuff–Related Shoulder Pain: A Systematic Review With Meta-analysis

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OBJECTIVE: To estimate the effects of rehabilitation interventions on strength in people with rotator cuff–related shoulder pain (RCRSP).

DESIGN: Intervention systematic review with meta-analysis.

LITERATURE SEARCH: Seven databases were searched from inception to April 2024.

STUDY SELECTION CRITERIA: Clinical trials or cohort studies reporting the effects of interventions lasting at least 6 weeks on objective strength outcomes in adults with RCRSP were included. Studies were excluded if they involved adults with other shoulder-related pathologies, focused exclusively on rotator cuff tears, or included participants who had undergone shoulder surgery.

DATA SYNTHESIS: We grouped strength testing results from individual studies according to the direction of the strength test. Meta-analyses were conducted using standardized within-group mean difference from baseline to primary follow-up for each study intervention arm.

RESULTS: Twenty-eight studies were included in the meta-analysis. Rehabilitation interventions had a moderate effect on shoulder external rotation strength (standardized mean difference [SMD], 0.56; 95% confidence interval [CI]: 0.39, 0.74; $I^2 = 78\%$; $P < .01$) and a small effect on shoulder abduction (SMD, 0.47; 95% CI: 0.32, 0.63; $I^2 = 54\%$; $P < .01$), internal rotation (SMD, 0.41; 95% CI: 0.30, 0.52; $I^2 = 25\%$; $P < .08$), flexion (SMD, 0.25; 95% CI: 0.01, 0.49; $I^2 = 0\%$; $P = .46$), and scaption (SMD, 0.48; 95% CI: 0.27, 0.70; $I^2 = 35\%$; $P = .14$) strength. Interventions that included strength training or active exercises had small-to-moderate effects on all strength outcomes. Interventions without exercises or strength training had no effects.

CONCLUSION: Rehabilitation interventions improved strength in people with RCRSP. Interventions that incorporated active exercises or strength training are recommended to address strength deficits in people with RCRSP. *J Orthop Sports Phys Ther 2026;56(2):67-84. Epub 16 December 2025. doi:10.2519/jospt.2025.13445*

KEY WORDS: *rotator cuff, shoulder, strength training, therapeutic exercise*

Shoulder pain has a global incidence ranging from 7.7 to 62 per 1000 people[35] and is one of the most common reasons why people seek primary care.[42] Among adults with shoulder pain, rotator cuff–related shoulder pain (RCRSP) is one of the most frequently reported conditions.[66] RCRSP is a comprehensive term that encompasses multiple shoulder disorders, including subacromial pain, subacromial impingement syndrome, rotator cuff tendinopathy, nonspecific shoulder pain, and rotator cuff tears.[33] Using this comprehensive terminology helps address the limitations of pathoanatomical diagnoses of shoulder pain, which attribute pain and functional limitations to specific structures.[33]

People with nonspecific shoulder pain frequently report limitations in strength-dependent activities (eg, lifting and carrying objects, exercise tolerance).[60] Shoulder weakness can also be associated with biomechanical alterations, such as altered scapular kinematics.[31] In patients with RCRSP, greater weakness is associated with higher levels of disability,[56] increased depression, and lower quality of life.[2] Therefore, addressing shoulder weakness is an essential goal of managing RCRSP.

Rehabilitation is a common approach for managing RCRSP, and clinical practice

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guidelines strongly recommend active exercises.[15] The primary goals of these exercises are to reduce pain and improve function and strength.[53,69] Active exercises can improve pain and function in the short, medium, and long terms.[32] However, previous reviews did not include shoulder strength as an outcome. There is substantial variability in the parameters used to prescribe active exercises for patients with RCRSP,[18] which may partly explain the inconsistent findings of clinical trials comparing the effect of different exercise protocols on shoulder strength in patients with RCRSP and healthy individuals.[43,58] As a result, the effect of active exercises on shoulder weakness and strength recovery remains unclear.

The primary aim of this systematic review was to estimate the within-group effect of rehabilitation interventions on shoulder strength in patients with RCRSP. As secondary aims, we investigated the within-group effect of the different types of strengthening interventions and the duration of the interventions on shoulder strength.

METHODS

This systematic review followed the Preferred Reporting Items for Systematic Reviews and Meta-Analyses guideline.[50] The protocol was prospectively registered in PROSPERO (CRD42024524534).

Search Strategy

A librarian from the University of Florida Health Science Center Libraries developed and conducted the searches in 7 electronic databases: PubMed (National Library of Medicine), EMBASE (Elsevier), Web of Science (Clarivate Analytics), the Cochrane Library (including the Cochrane Database of Systematic Reviews and the Cochrane Central Register of Controlled Trials), SPORTDiscus (EBSCOhost), CINAHL (EBSCOhost), and PEDro. The librarian developed a pilot search in PubMed to identify key articles included in previous systematic reviews[32] and clinical practice guidelines.[15] The librarian used this pilot search to adjust the key words and subject headings used in the final PubMed search strategy, which was then translated to the other databases. The time range for the final search started from database inception to April 12-16, 2024. The complete search strategy for PubMed is reported in **SUPPLEMENTAL TABLE 1**. The reference lists of the included studies were also hand-searched for additional missing studies.

Study Selection

The following inclusion criteria were used to determine eligibility: (1) the inclusion of patients with RCRSP aged 18 years or older; (2) objective measurement of strength using a dynamometer; (3) testing of a nonsurgical intervention lasting at least 6 weeks; (4) a randomized controlled trial or a nonrandomized prospective cohort design; and (5) publication in English, Spanish, Italian, French, Chinese, or Portuguese. The 6-week intervention duration was chosen based on guideline recommendations.[47,51] The exclusion criteria were as follows: (1) the inclusion of patients with other shoulder-related pathologies (ie, shoulder instability, frozen shoulder, labrum tear, shoulder fractures, biceps tendinopathy, thoracic outlet syndrome, coracoid impingement syndrome, neck radiculopathy); (2) the inclusion of patients who underwent shoulder surgery; and (3) case studies involving fewer than 10 patients, narrative reviews, guidelines, consensus statements, editorials, letters, comments, and conference abstracts. Although RCRSP encompasses rotator cuff tears, we excluded studies with only participants with full-thickness rotator cuff tears due to potentially different mechanisms of strength recovery in this population.

Covidence systematic review software (Veritas Health Innovation, Melbourne, Australia) was used to remove duplicates and screen titles, abstracts, and full-text articles. Two authors independently screened titles and abstracts, followed by full-text screening. Any disagreements were resolved via consensus. A third reviewer was consulted to resolve conflicts at each step (abstract/title or full-text screening) and make a final decision if consensus could not be reached.

Data Extraction

Two authors independently extracted data from each included study. Any discrepancies were discussed and resolved via consensus. If consensus was not achieved, a third author resolved conflicts. The following information was extracted: (1) general information (ie, authors' names, year, nationality, design of the study, sample size population), (2) demographic variables (ie, sex, age, duration of symptoms), (3) intervention characteristics (ie, procedures, material, location, target, frequency, session duration, length, dose), (4) strength measurement (ie, movement, tool, contraction type, descriptive metrics), and (5) baseline and follow-up strength outcomes.

Risk of Bias

Two reviewers independently assessed the risk of bias for all the included studies. Discrepancies were discussed and resolved via consensus. A third reviewer was consulted if consensus was not reached. The Newcastle-Ottawa Quality Assessment Scale[49] was used to evaluate the risk of bias in nonrandomized trials and prospective studies in 3 domains: selection, comparability, and outcomes. After scoring each domain, the study was classified as "Good", "Fair", or "Poor".[49] The revised Cochrane risk-of-bias tool (RoB2)[63] was used to evaluate the risk of bias in randomized trials. RoB2 evaluates different domains related to study design, conduct, and reporting. Each domain was weighted, and an extent of bias risk was proposed for each domain separately, along with a final score. Thus, RoB2 categorizes studies as having "Low" risk of bias, "High" risk of bias, or "Some concerns".[63]

Outcome Measure

Strength was the primary outcome of interest. We extracted strength test

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characteristics, including assessment tool, movement direction, contraction type, movement speed, number of trials, body position, arm position, resistance location, and normalization parameters. For studies including multiple testing locations for the same direction (eg, isometric strength at different degrees of humeral elevation), we selected the most common position across studies. For studies using an electromechanical dynamometer, we included isokinetic testing at the lowest tested speed and excluded eccentric strength testing.

We extracted raw or normalized strength values from baseline to posttreatment end points. The end point at the conclusion of the intervention was considered the primary end point for the systematic review.

Intervention

Two reviewers extracted information regarding the intervention content for each study arm based on the categories of the Template for Intervention Description and Replication.[24] Each study arm was then categorized according to its intervention characteristic using a consensus approach. The strengthening group included arms primarily focused on muscle-strengthening exercises. The multimodal group consisted of arms focused on conventional therapy or standard care, which may have involved active rehabilitation exercises but were not specifically designed to enhance muscle strength. The education group included arms focused solely on education. The sham/placebo included arms that involved a sham or placebo exercise therapy. The natural course group included arms with no intervention, following patients longitudinally.

A secondary classification subdivided the strengthening group based on the specific type of strengthening interventions. These subcategories included nonspecific, motor control, concentric, eccentric, neuromuscular electrical stimulation (NMES), high load, low load, blood flow restriction, open kinetic chain, closed kinetic chain, and unaffected area (strength training of an area away from the affected shoulder).

Statistical Analysis

All analyses were performed using Stata (version 16, StataCorp LLC). Means, standard deviations (SDs), and confidence intervals (CIs) for each study arm were extracted independently. If a study reported data with means and CIs or standard errors, SDs were extrapolated based on methods outlined in the Cochrane Handbook.[11] Medians and interquartile ranges were excluded from the meta-analysis. When the data were not reported, the corresponding authors were contacted to obtain relevant data.

We grouped strength testing results from individual studies according to the direction of the strength test (external rotation, internal rotation, scaption [elevation in the scapular plane], abduction, adduction, flexion, and extension). We used the within-group mean difference from baseline to primary follow-up to quantify the effects of intervention types on shoulder strength. This approach aligns with our study aims, allows for the inclusion of randomized controlled trials and longitudinal cohort studies, and increases available data for a more precise estimate of intervention effects. We calculated the standardized mean difference (SMD) between baseline and primary follow-up for each study arm, which allows combining data despite variability in strength testing methods, outcomes, and units of measurement. When data were missing, posttreatment values were extrapolated using the difference between baseline and mean change values, with the SD imputed based on the baseline SD.

We performed separate meta-analyses for each strength test movement direction. In the primary meta-analyses, study arms were subgrouped by the type of intervention (strengthening, multimodal, education, sham/placebo, natural course). We also performed 2 secondary meta-analyses. First, we took the study arms classified in the strengthening group from the primary meta-analyses and subgrouped them based on the type of strengthening intervention (nonspecific, motor control, concentric, eccentric, NMES, high load, low load, blood flow restriction, unaffected area). Second, we took study arms that used active interventions (excluding education and natural course groups) and subgrouped them by the length of the intervention (6, 8, 9, and 12 weeks). The secondary meta-analyses only included strength test directions that were measured in at least 10 study arms to have a more reliable representation of the effect.

For all analyses, we calculated the overall effect and individual effects for each subgroup. The random-effects model with the inverse-variance method was applied to estimate the weighted mean SMD, and Cohen's *d* was used to interpret effect sizes as small (0.00-0.49), moderate (0.5-0.79), or large (0.8). The I^2 statistic was used to evaluate heterogeneity, which was interpreted as low for I^2 values of 0%-25%; moderate, 25%-75%; and high, 75%-100%.[23]

Sensitivity Analysis

We performed sensitivity analyses to assess the robustness of the primary meta-analysis findings by excluding the studies that we classified as having high risk of bias.

RESULTS

The search identified 6659 studies (**FIGURE**). After removing 3399 duplicates, 3260 studies were screened, and 280 were reviewed in full text. Two studies[27,65] from the same authors overlapped in design, methodology, and results; the study with the older publication date[27] was included in the systematic review. Thirty-three studies[1,3–10,12–14,16,17,19–21,25–30,34,36,39,45,46,48,54,57,64,71] met the eligibility criteria, with 28 studies[1,3–7,9,10,12–14,16,17,20,21,25–30,34,36,39,46,54,64,71]

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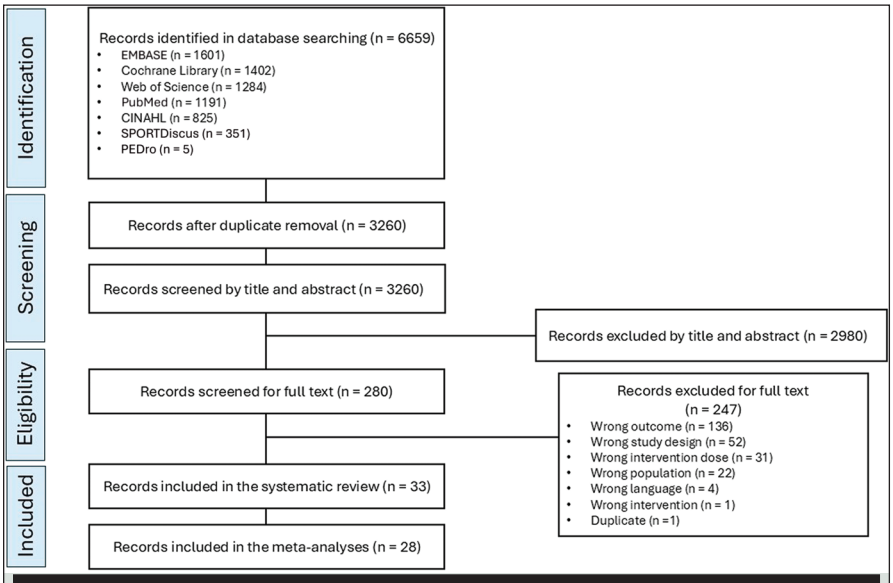


FIGURE. Preferred Reporting Items for Systematic Reviews and Meta-Analyses flow diagram.

included in the meta-analysis. Five studies were excluded from the meta-analysis for different reasons, including incomplete strength outcomes,[8,45,48] using median and interquartile range for results reporting,[19] and reporting data for the right and left sides without specifying the affected side.[57]

Study Characteristics

TABLE 1 presents the general information of the studies included. Thirty studies[1,3–10,12–14,16,17,19–21,25–30,34,36,48,54,57,64,71] (91%) were randomized controlled trials, 2 studies[39,46] (6%) were prospective studies, and 1 study[45] (3%) was a nonrandomized controlled trial. This systematic review comprised studies from 17 countries (55% high income, 39% upper middle income, 6% lower middle income; **SUPPLEMENTAL TABLE 2**).

We included data from 1702 individuals with RCRSP; female participants accounted for 49% of the sample (n = 831). None of the included studies reported participant race or ethnicity. Seventeen studies[3–5,7–9,12,16,20,29,34,36,39,46,48,54,64] (52%) used the diagnostic label of subacromial impingement syndrome; 6 studies[10,14,19,25,30,57] (18%), subacromial pain syndrome; 5 studies[1,13,21,26,28] (15%), rotator cuff tendinopathy; 2 studies[27,45] (6%), shoulder pain; 1 study[17] (3%), RCRSP; 1 study[6] (3%), rotator cuff disease; and 1 study[71] (3%), shoulder disorder. Fourteen studies[1,6,7,10,14,17,20,21,25,26,28,30,39,54] (42%) excluded participants with full-thickness rotator cuff tears, whereas 5 studies[8,13,19,36,48] (15%) excluded participants with full- and partial-thickness rotator cuff tears. Thirteen studies[3–5,9,12,27,29,34,45,46,57,64,71] (39%) did not explicitly state whether participants with rotator cuff tears were excluded. Three studies[16,29,53] reported the number of included participants with rotator cuff tears (n = 36).[16,29,54] Eighteen studies[1,3–7,9,10,12–14,16,17,19,20,25–28,30,34,36,46,48,71] (55%) reported the average duration of symptoms, which ranged between 4 months and 3.6 years. Seven studies[3,4,14,19,20,36,46] (21%) used duration of symptoms as an inclusion criterion for eligibility.

Risk of Bias

Eight studies[6,12,17,25,26,28,34,64] (27%) were at low risk of bias, 9 studies[1,7,9,10,13,14,16,20,30,71] (30%) had some concerns, and 13 studies[3,4,8,9,19,27,29,36,48,54,57] (43%) were at high risk of bias (**SUPPLEMENTAL FIGURE 1**). For the Newcastle-Ottawa Quality Assessment Scale, all the studies were classified as poor (**SUPPLEMENTAL TABLE 3**).

Study Interventions

TABLE 2 presents the intervention characteristics of all the included studies. The intervention duration ranged between 6 and 12 weeks. Intervention characteristics varied across studies. Most of the studies used individual (79%) sessions, with a frequency of 3 (42%) or 2 (27%) times per week and session durations ranging from 20 to 120 minutes. Home exercise programs were included in 18 studies[1,3,6,7,10,12,13,16,17,19,26,27,29,30,36,39,54,64] (55%). Exercise dosage varied across studies: the number of sets ranged from 1 to 12 (reported in 79% of studies), and repetitions ranged from 5 to 30 (reported in 89% of studies). Twenty-seven studies[1,3–5,10,13,16,17,19,20,25–30,34,36,39,45,46,48,54,57,64,71] reported on the progression of exercises during the intervention period.

For the primary analysis, 42 arms were classified as strengthening; 20 arms, as multimodal; 5 arms, as natural course; 1 arm, as sham/placebo; and 1 arm, as education. For the secondary analyses, we categorized the strengthening arms by type: nonspecific (11 arms); motor control (9 arms); eccentric (8 arms); concentric (3 arms); unaffected area, device, and NMES (2 arms each); and open kinetic chain, closed kinetic chain, blood flow restriction, high load, and low load (1 arm each). We also grouped the arms by intervention duration: 6 weeks (16 arms), 8 weeks (21 arms), 9 weeks (2 arms), 10 weeks (2 arms), and 12 weeks (19 arms).

Strength Outcomes

TABLE 3 summarizes the strength testing parameters. Twenty-seven studies[1,3,5–8,10,12–14,17,19,20,25–27,29,30,36,39,45,46,48,57,64,71] (82%) measured isometric strength using a handheld dynamometer. Six studies[4,9,16,21,28,34] (18%) used an isokinetic dynamometer to measure either isometric[16] (1 study), concentric[4,21,28,34] (4 studies), or eccentric[4] (1 study) strength. The isokinetic test speeds ranged from 60° to 300° per second. Strength test movement direction included external rotation (31 studies),[1,3–10,12,14,16,17,19,21,25–30,34,36,39,45,46,48,54,57,71]

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TABLE 1 Study Characteristics

Author (Year)	Design	Rotator Cuff Tear Exclusion Criteria	Arm	Sample Size	Included With Rotator Cuff Tears[a]	Sex (Female)[b]	Age	Symptom Duration (Months)	Dropout[b]	Primary End Point (Weeks)
Ager et al (2019)[1]	RCT	Full tears	1	16	NR	0 (0)	33.4 (9.5)	23.2 (41.5)	1 (6)	6
			2	15		1 (7)	36.9 (7.1)	38.9 (50.5)	3 (20)	
AlAnazi et al (2022)[3]	RCT	NR	1	20	NR	14 (78)	39.1 (8.5)	<12[c]	2 (10)	8
			2	20		12 (75)	39.2 (7.6)		4 (20)	
de Oliveira et al (2022)[14]	RCT	Full tears	1	12	NR	9 (75)	46.2 (8.1)	>2[c]	1 (8)	8
			2	12		9 (75)	48.5 (9.3)		1 (8)	
Babaei-Mobarakeh et al (2018)[4]	RCT	NR	1	15	NR	NR	25.5 (1.05)	>3[c]	0	8
			2	15			28.2 (2.02)		0	
Başkurt et al (2011)[5]	RCT	NR	1	20	NR	NR	51.6 (8.4)	8.6 (10.8)	0	6
			2	20			51.3 (11.6)	11.6 (9.5)	0	
Bennell et al (2010)[6]	RCT	Full tears	1	59	NR	26 (42)	59.3 (10.1)	24 (6-54)[d]	3 (5)	11
			2	61		31 (51)	60.8 (12.4)	14 (6-24)[d]	3 (5)	
Blume et al (2015)[7]	RCT	Full tears	1	18	NR	10 (56)	50.1 (16.9)	28.2 (23.6)	0	8
			2	20		10 (63)	48.6 (14.6)	20.6 (26.6)	4 (20)	
Boeck et al (2012)[8e]	RCT	Full or partial tears	1	7	NA	7 (100)	54.7 (12.8)	NR	NR	8
			2	7		6 (86)	56.4 (11.7)			
Cha et al (2014)[9]	RCT	NR	1	16	NR	0 (0)	21.3 (1.7)	6.1 (1.4)	0	12
			2	14		0 (0)	22.6 (1.8)	5.6 (1.2)	0	
Chaconas et al (2017)[10]	RCT	Full tears	1	25	NR	10 (40)	43.4 (17.9)	53.4 (84.7)	3 (12)	6
			2	23		10 (44)	48.4 (16.9)	44.9 (75.5)	9 (39)	
Dejaco et al (2016)[13]	RCT	Full or partial tears	1	20	NA	10 (50)	50.2 (10.8)	16.9 (16.8)	1 (5)	12
			2	16		7 (44)	48.6 (12.3)	23.1 (23.8)	1 (6)	
Dilek et al (2016)[16]	RCT	NR	1	31	PT = 17 (28%) FT = 13 (23%)	23 (74)	50.1 (10.83)	17.0 (26.2)	1 (3)	12
			2	30		19 (63)	48.2 (9.74)	17.1 (24.6)	1 (3)	
Dubé et al (2023)[17]	RCT	Full tears	1	41	NR	20 (49)	47.9 (15.3)	24 (54.0)	5 (12)	12
			2	41		21 (51)	48.4 (14.6)	24 (32.5)	4 (10)	
			3	41		22 (54)	46.9 (15.7)	18 (47.0)	5 (12)	
Ellegaard et al (2016)[19e]	RCT	Full or partial tears	1	49	NA	30 (61)	49.4 (13.1)	>1[c]	18 (36)	13
			2	50		28 (56)	47.7 (13.3)		18 (36)	
Elnour et al (2023)[20]	RCT	Full tears	1	17	NR	11 (65)	32.9 (5.3)	>3[c]	0	6
			2	17		12 (71)	31.2 (6.0)		0	
Ersever et al (2022)[21]	RCT	Full tears	1	21	NR	18 (86)	52 (6.7)	NR	0	6
			2	20		9 (45)	45.2 (9.7)		0	
de Freitas et al (2014)[12]	RCT	NR	1	26	NR	16 (62)	50.1 (8.2)	22 (17.7)	4 (15)	9
			2	30		20 (67)	50.8 (9.6)	21.2 (19.0)	6 (20)	
Hotta et al (2020)[25]	RCT	Full tears	1	30	NR	21 (70)	47 (10)	28 (15)	3 (10)	8
			2	30		21 (70)	51 (8)	29 (33)	4 (13)	
Ingwersen et al (2017)[26]	RCT	Full tears	1	49	NR	19 (39)	45.7 (10.6)	12 (3-180)[d]	5 (10)	12
			2	51		27 (53)	46.5 (10.1)	12 (3-180)[d]	2 (4)	
Kanthanathan et al (2023)[27]	RCT	NR	1	63	NR	21 (46)	45.0 (10.3)	4.06 (3.6)	17 (27)	12
			2	63		21 (41)	45.1 (11.4)	4.53 (3.2)	12 (19)	
Kara et al (2024)[28]	RCT	Full tears	1	14	PT = 5 (17.8%)	7 (50)	31.1 (28.2-34.1)[f]	7.5 (5-12)[d]	0	8
			2	14		7 (50)	34.9 (31.7-38.1)[f]	7 (4-8)[d]	0	

(Table continues on next page.)

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TABLE 1 Study Characteristics (continued)

Author (Year)	Design	Rotator Cuff Tear Exclusion Criteria	Arm	Sample Size	Included With Rotator Cuff Tears[a]	Sex (Female)[b]	Age	Symptom Duration (Months)	Dropout[b]	Primary End Point (Weeks)
Karaaslan et al (2023)[29]	RCT	NR	1	25	NR	15 (60)	43.5 (9.2)	NR	1 (4)	8
			2	25		20 (80)	48.3 (9.1)		1 (4)	
Karabay et al (2025)[30]	RCT	Full tears	1	15	NR	11 (73)	43.9 (14.3)	6 (3-8)[d]	3 (25)	12
			2	15		11 (73)	47.5 (15.1)	5 (3-6)[d]	2 (13)	
			3	15		10 (67)	48.1 (14.8)	7 (4-12)[d]	3 (25)	
Lombardi et al (2008)[34]	RCT	NR	1	30	NR	21 (70)	56.3 (11.6)	13.7 (9.6)	0	8
			2	30		25 (83)	54.8 (9.4)	13.9 (9.3)	4 (13)	
Maenhout et al (2013)[36]	RCT	Full or partial tears	1	31	NA	16 (52)	40.2 (12.9)	>3[d]	3 (10)	12
			2	30		20 (67)	39.4 (13.1)		8 (27)	
McClure et al (2004)[39]	PLC	Full tears	1	39	NR	21 (54)	50.6 (13.1)	NR	20 (34)	6
Nyawose & Naidoo (2021)[45e]	Non-RCT	NR	1	15	NR	13 (87)	41.2 (7.5)	NR	NR	8
			2	15		15 (100)	48.6 (7.6)			
Ortega-Cebrián et al (2021)[46]	PLC	NR	1	34	NR	0 (0)	29.1 (5.5)	>3[d]	8 (23)	12
Østerås & Torstensen (2010)[48e]	RCT	Full or partial tears	1	31	NA	7 (22)	46.1 (11.2)	3.6 (5.1)[g]	4 (13)	12
			2	30		6 (19)	41.8 (14.5)	3.1 (4.3)[g]	5 (17)	
Rizzo et al (2017)[54]	RCT	Full tears	1	8	PT = 1 (7.1%)	NR	NR	NR	0	6
			2	6					3 (50)	
Schedler et al (2020)[57e]	RCT	NR	1	24	NR	8 (47)	52.7 (7.0)	NR	7 (29)	8
			2	23		10 (53)	54.3 (8.4)		4 (17)	
			3	22		11 (55)	53.0 (6.6)		2 (9)	
Struyf et al (2013)[64]	RCT	NR	1	12	NR	5 (50)	45.4 (15.1)	NR	2 (17)	4-8
			2	10		7 (58)	46.2 (13.5)		0	
Wang & Trudelle-Jackson (2006)[71]	RCT	NR	1	20	NR	9 (60)	39.4 (13.2)	49.7 (52.4)[h]	5 (25)	8
			2	18		6 (40)	49.9 (18.3)	41.7 (51.5)[h]	3 (17)	

All data are presented as mean (SD) unless otherwise indicated.
Abbreviations: FT, full-thickness rotator cuff tears; NA, not applicable; NR, not reported; PLC, prospective longitudinal cohort; primary end point, end of intervention; PT, partial-thickness rotator cuff tears; RCT, randomized controlled trial.
[a]*Number and % of total study sample size.*
[b]*Number and %.*
[c]*Duration of symptoms that were part of the inclusion criteria.*
[d]*Median and interquartile range.*
[e]*Not included in the quantitative analysis.*
[f]*Mean and 95% confidence interval.*
[g]*Years.*
[h]*Weeks.*

internal rotation (25 studies),[3–6,8–10,12,19–21,25,26,28,29,34,36,39,45,46,48,54,57,71] abduction (18 studies),[1,5–8,10,13,16,17,19,25,27,29,34,45,48,71] scaption (6 studies),[12,26,30,36,39,64] flexion (6 studies),[8,14,29,34,45,48] extension (2 studies),[14,34] and adduction (1 study).[34] The strength testing procedures varied in terms of the number of trials performed (ranging 2-10 trials; 12 studies[1,4,8,9,20,25,34,39,45,46,48,64] did not report this information), outcome selection (14 studies[3,5–7,12–14,16,26,29,30,36,54,71] used between-trials average, 7 studies[10,17,19,21,28,34,57] reported the peak value, 13 studies[1,4,8,9,20,25,27,39,45,46,48,64] did not specify which value was used), and reported normalization parameters (reported in 6 trials[10,12,17,25,26,28]). Other strength testing parameters were participant positioning, arm position, and resistance placement (**SUPPLEMENTAL TABLE 4**).

Meta-analyses

The primary meta-analysis included 28 studies, 56 arms, and 1429 patients (**TABLE 4**). When combining all rehabilitation interventions, a significant moderate effect was observed favoring

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TABLE 2 Intervention Characteristics

		Systematic Review Classification													
Author (Year)	Arm	Primary	Secondary	Provider	Cointervention	Targets	Setting	Type	HEP	Session[a]	Duration[b]	Length[c]	Sets	Reps	Progression
Ager et al (2019)[1]	1	Strengthening	Motor control	PT	None	RC, SC	MF	Group	No	3	35-45	6	NR	NR	Yes
	2	Multimodal		PT	CP, ED, taping, MT	RC	MF	Individual	Yes	2-3	30	6	NA	NA	NR
AlAnazi et al (2022)[3]	1	Strengthening	Unaffected area	PT	CT	Forearm	MF	Individual	Yes	2	NR	8	3	10	Yes
	2	Multimodal		PT	None	NR	MF	Individual	Yes	2	NR	8	NA	NA	NA
de Oliveira et al (2022)[14]	1	Strengthening	Motor control	NR	None	RC, SC	Research center	Individual	NR	2	40	8	3	8-15	NR
	2	Multimodal		NR	None	RC, SC	Research center	Individual	NR	2	40	8	3	8-15	NR
Babaei-Mobarakeh et al (2018)[4]	1	Strengthening	Device	PT	None	RC	NR	Individual	NR	3	30	8	3-6	30-60[d]	Yes
	2	Natural course		NA	None	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
Başkurt et al (2011)[5]	1	Strengthening	Motor control	PT	ED, ST	RC, SC	NR	Individual	NR	3	NR	6	3	10	Yes
	2	Strengthening	Nonspecific	PT	ED, ST	RC	NR	Individual	NR	3	NR	6	3	10	Yes
Bennell et al (2010)[6]	1	Multimodal		PT	Pain medication	RC, SC	MF	Individual	Yes	Various[f]	30-45	12	5	15	NR
	2	Sham/placebo[e]		PT	Pain medication	NA	MF	Individual	NR	Various[f]	30-45	12	NA	NA	NA
Blume et al (2015)[7]	1	Strengthening	Eccentric	PT	NR	RC	MF	Individual	Yes	2	NR	8	1-3	12-20	Yes
	2	Strengthening	Concentric	PT	NR	RC	MF	Individual	Yes	2	NR	8	1-3	12-20	Yes
Boeck et al (2012)[8]	1	Strengthening	CKC	NR	NR	NR	NR	NR	NR	3	50	8	NR	NR	NR
	2	Strengthening	OKC	NR	NR	NR	NR	NR	NR	3	50	8	NR	20	NR
Cha et al (2014)[9]	1	Multimodal		NR	US, laser, ET, AC	RC	MF	Individual	NR	3	90-120	12	5-12	10-15	NR
	2	Natural course		NA	NA	NA	NA	NA	NR	NA	NA	NA	NA	NA	NA
Chaconas et al (2017)[10]	1	Strengthening	Eccentric	PT	NR	RC	MF	Individual	Yes	Various[g]	NR	6	2-3	10-15	Yes
	2	Multimodal		PT	NR	RC, SC	MF	Individual	Yes	Various[h]	NR	6	NR	NR	NR
Dejaco et al (2016)[13]	1	Strengthening	Eccentric	PT	ST	RC	MF	Individual	Yes	Various[i]	NR	12	3	8	Yes
	2	Strengthening	Nonspecific	PT	ST	RC	MF	Individual	Yes	Various[i]	NR	12	3	8	Yes
Dilek et al (2016)[16]	1	Strengthening	Motor control	Researcher	Pain medication	RC, SC	MF	Individual	Yes	3	NR	12	NR	NR	Yes
	2	Multimodal		Researcher	Pain medication	RC, SC	MF	NR	Yes	3	NR	12	NR	NR	NA
Dubé et al (2023)[17]	1	Education		PT	ED	NA	Research center	Individual	Yes	NA	NA	NA	NA	NA	NA
	2	Strengthening	Nonspecific	PT	ED	RC, SC	Research center	Individual	Yes	Various[j]	30	12	1	Fatigue	Yes
	3	Strengthening	Motor control	PT	Injection	RC, SC	Research center	Individual	Yes	Various[j]	30	12	1-3	15	Yes
Ellegaard et al (2016)[19]	1	Strengthening	Motor control	PT	Injection	RC, SC	MF	Individual	Yes	3	NR	10	NR	NR	Yes
	2	Strengthening	Unaffected area[k]	PT	ST	RC, SC	MF	Individual	Yes	3	NR	10	NR	NR	Yes
Elnour et al (2023)[20]	1	Strengthening	Motor control	PT	None	RC, SC	MF	NR	NR	3	35-45	6	3	10	Yes
	2	Strengthening	Nonspecific	PT	CP/HP, MT, ST	RC, SC	MF	NR	NR	3	NR	6	1	10	Yes

(Table continues on next page.)

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TABLE 2 INTERVENTION CHARACTERISTICS (CONTINUED)

		Systematic Review Classification													
Author (Year)	Arm	Primary	Secondary	Provider	Cointervention	Targets	Setting	Type	HEP	Session[a]	Duration[b]	Length[c]	Sets	Reps	Progression
Ersever et al (2022)[21]	1	Multimodal		PT	PEMF, ST	RC, SC	MF	NR	NR	3	NR	6	NR	NR	NR
	2	Strengthening	NMES	PT		RC, SC	MF	NR	NR	3	NR	6	NR	NR	NR
de Freitas et al (2014)[12]	1	Multimodal		PT	Placebo PEMF, ST	RC, SC	MF	Individual	Yes	2	NR	9	3	10	NR
	2	Multimodal		PT	None	RC, SC	MF	Individual	Yes	2	NR	9	3	10	NR
Hotta et al (2020)[25]	1	Strengthening	Nonspecific	PT	None	SC	MF	Individual	NR	3	50	8	3	10	Yes
	2	Strengthening	Motor control	PT	Injection (if needed)	SC	MF	Individual	NR	3	50	8	3	10	Yes
Ingwersen et al (2017)[26]	1	Strengthening	High load	PT	MT, ST	RC, SC	MF	Individual	Yes	3	30	12	3-4	6-15	Yes
	2	Strengthening	Low load	PT	MT, ST	RC, SC	MF	Individual	Yes	3	30	12	3	20-25	Yes
Kanthanathan et al (2023)[27]	1	Strengthening	Eccentric	PT	MT, ST	RC, SC	MF + home	Individual	Yes	Various[l]	NR	12	3	15	Yes
	2	Strengthening	Concentric	PT	ED, CD, ST	RC, SC	MF + home	Individual	Yes	Various[l]	NR	12	3	15	Yes
Kara et al (2024)[28]	1	Strengthening	BFR	PT	Self-assisted ROM	RC, SC	MF	Individual	NR	2	NR	8	4	Various[m]	Yes
	2	Strengthening	Nonspecific	PT	Self-assisted ROM	RC, SC	MF	Individual	NR	2	NR	8	4	Various[m]	Yes
Karaaslan et al (2023)[29]	1	Strengthening	Nonspecific	PT	TENS, HP, ST, ROM	RC, SC	MF	Individual	Yes	2	NR	8	1	10	Yes
	2	Strengthening	NMES	PT	TENS, HP, ST, ROM	RC, SC	MF	Individual	Yes	2	NR	8	1	10	Yes
Karabay et al (2025)[30]	1	Strengthening	Eccentric	PT	Pain medication	RC	MF	Individual	Yes	3.5	NR	12	3	15	Yes
	2	Strengthening	Concentric	PT	Pain medication	RC	MF	Individual	Yes	Various[n]	NR	12	3	15	Yes
	3	Multimodal		PT	Pain medication	NA	MF	Individual	Yes	Various[n]	NR	12	NA	NA	NA
Lombardi et al (2008)[34]	1	Strengthening	Nonspecific	PT	None	RC	NR	NR	NR	2	NR	8	2	8	Yes
	2	Natural course		NA	None	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
Maenhout et al (2013)[36]	1	Strengthening	Eccentric	PT	None	RC	MF + home	Individual	Yes	Various[o]	30	12	3	15	Yes
	2	Strengthening	Nonspecific	PT	ST, ROM	RC	MF + home	Individual	Yes	Various[o]	30	12	3	10	Yes
McClure et al (2004)[39]	1	Multimodal		PT	ST, ROM	RC, SC	MF + home	Individual	Yes	Various[p]	NR	6	3	10	Yes
Nyawose & Naidoo (2021)[45]	1	Multimodal		PT	ST, ROM	RC, biceps	Work	NR	NR	2	25-30	8	2-3	10-12	Yes
	2	Natural course		NA	NA	NA	NA	NR	NA	NA	NA	NA	NA	NA	NA
Ortega-Cebrián et al (2021)[46]	1	Multimodal		PT	MT, ST	NR	MF	NR	NR	3	NR	12	NR	NR	Yes
Østerås & Torstensen (2010)[48]	1	Multimodal	High dosage	PT	Aerobic	NR	MF	Individual	NR	3	60	12	3	30	Yes
	2	Multimodal	Low dosage	PT	Aerobic	NR	MF	Individual	NR	3	30	12	2	10	Yes

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Rizzo et al (2017)[54]	1	Multimodal	OT	None	RC, SC	MF	Individual	Yes	2	30-40	6	NR	NR	Yes
	2	Multimodal	Researcher	None	RC, SC	MF	Individual	Yes	2	30-40	6	NR	NR	NA
Schedler et al (2020)[57]	1	Strengthening	Device	NR	RC	Home	Individual	NR	3	20–25	8	7	6-19	Yes
	2	Strengthening	Nonspecific	NR	RC	Home	Individual	NR	3	20–25	8	NR	NR	Yes
	3	Natural course		NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
Struyf et al (2013)[64]	1	Strengthening	Motor control	PT	MT, ST	RC, SC	MF	Individual	Yes	1-3	30	4-8	1	10	Yes
	2	Strengthening	Eccentric	PT	MT, US	RC, SC	MF	Individual	Yes	1-3	30	4-8	3	15	Yes
Wang & Trudelle-Jackson (2006)[71]	1	Multimodal	MI	PT	NR	RC, SC	NR	Individual	NR	10	NR	8	1	5	Yes
	2	Strengthening	Nonspecific	PT	NA	RC, SC	NR	Individual	NR	10	NR	8	1	5	Yes

Abbreviations: AC, air compression; BFR, blood flow restriction; CKC, closed kinetic chain; CP, cold pack; ED, education; ET, electrotherapy; HEP, home exercise program; HP, hot pack; MF, medical facility; MI, match with impairment; MT, manual therapy; NA, not applicable; NMES, neuromuscular electrical stimulation; NR, not reported; OKC, open kinetic chain; OT, occupational therapist; PEMF, pulsed electromagnetic field; PT, physical therapist; RC, rotator cuff muscles; ROM, range of motion; SC, scapular muscles; ST, stretching; TENS, transcutaneous electrical nerve stimulation; US, ultrasound.

[a] *Per week.*
[b] *Minutes.*
[c] *Weeks.*
[d] *Time in seconds.*
[e] *Inactive ultrasound therapy.*
[f] *Two sessions per week for 2 weeks, 1 session per week for 4 weeks, 1 session biweekly for 4 weeks.*
[g] *Eight sessions per week for weeks 1-3, 1 session per week for weeks 4 and 6.*
[h] *Eight sessions per week for weeks 1-3, 2 sessions per week for weeks 4 and 6.*
[i] *One session per week for 6 weeks, 3 sessions in the last 6 weeks, 7 sessions per week for home-based program.*
[j] *One session per week or biweekly.*
[k] *Contralateral rotator cuff scapular muscles.*
[l] *Three sessions of in-person intervention and not reported session numbers of home intervention per week.*
[m] *Thirty repetitions for the first set, 15 repetitions for the rest 3 sets.*
[n] *Three sessions per week for 6 weeks, 2 sessions per week for 3 weeks, 1 session per week for 3 weeks.*
[o] *One session per week in the first 6 weeks, 1 session biweekly in the second 6 weeks.*
[p] *One session of in-person intervention and 6 sessions of home intervention per week.*

postintervention outcomes for external rotation strength (49 arms; SMD, 0.56; 95% CI: 0.39, 0.74; I^2 = 78%; P < .01) (**SUPPLEMENTAL FIGURE 2**). There were significant small effects for postintervention outcomes for abduction (25 arms; SMD, 0.47; 95% CI: 0.32, 0.63; I^2 = 54%; P < .01) (**SUPPLEMENTAL FIGURE 3**), internal rotation (37 arms; SMD, 0.41; 95% CI: 0.30, 0.52; I^2 = 25%; P < .08) (**SUPPLEMENTAL FIGURE 4**), flexion (6 arms; SMD, 0.25; 95% CI: 0.01, 0.49; I^2 = 0%; P = .46) (**SUPPLEMENTAL FIGURE 5**), and scaption (9 arms; SMD, 0.48; 95% CI: 0.27, 0.70; I^2 = 35%; P = .14) (**SUPPLEMENTAL FIGURE 6**) strength. There was no effect for extension strength (4 arms; SMD, 0.05; 95% CI: −0.25, 0.36; I^2 = 0%; P = .14) (**SUPPLEMENTAL FIGURE 7**).

Strengthening interventions had a moderate effect for external rotation (29 arms; SMD, 0.64; 95% CI: 0.43, 0.86; I^2 = 76%; P < .01), abduction (18 arms; SMD, 0.58; 95% CI: 0.40, 0.77; I^2 = 53%; P < .01), and scaption (6 arms; SMD, 0.58; 95% CI: 0.26, 0.91; I^2 = 52%; P = .06) strength and a small effect for internal rotation (21 arms; SMD, 0.47; 95% CI: 0.34, 0.59; I^2 = 0%; P < .85) and flexion (4 arms; SMD, 0.38; 95% CI: 0.08, 0.67; I^2 = 0%; P < .48) strength. Multimodal interventions had a moderate effect for external rotation strength (16 arms; SMD, 0.57; 95% CI: 0.23, 0.92; I^2 = 80%; P < .01); a small effect for abduction (4 arms; SMD, 0.29; 95% CI: 0.03, 0.55; I^2 = 0%; P < .89), internal rotation (13 arms; SMD, 0.46; 95% CI: 0.22, 0.69; I^2 = 48%; P < .05), and scaption (3 arms; SMD, 0.31; 95% CI: 0.03, 0.60; I^2 = 0%; P < .96) strength; and no effect for flexion strength (1 arm; SMD, 0.03; 95% CI: −0.77, 0.83). Education (1 arm from 1 study), sham/placebo (1 arm from 1 study), and natural course (up to 2 arms from 2 studies) showed no effect on strength outcomes. No significant effect was found for adduction (2 arms from 1 study) and extension (4 arms from 2 studies) in any group classification.

Sensitivity Analysis Excluding studies with high risk of bias or rated as poor quality (n = 11, 39%) resulted in

TABLE 3 Strength Testing Characteristics

Author (Year)	Strength Assessment Tool	Movement Speed	Contraction Type	Number of Trials	Outcome	Normalization Parameter	Movement Direction						
							ABD	ER	IR	FLX	SCAP	EXT	ADD
Ager et al (2019)[1]	HHD	NA	Isometric	NR	NR	NR	X	X					
AlAnazi et al (2022)[3]	HHD	NA	Isometric	2	Average	NR		X	X				
de Oliveira et al (2022)[14]	HHD	NA	Isometric	2	Average	NR		X	X	X		X	
Babaei-Mobarakeh et al (2018)[4]	Isokinetic dynamometer	180°/s[a] 300°/s	Concentric[a] Eccentric	NR	NR	NR		X	X				
Başkurt et al (2011)[5]	HHD	NA	Isometric	3	Average	NR	X	X	X				
Bennell et al (2010)[6]	HHD	NA	Isometric	4	Average	NR	X	X	X				
Blume et al (2015)[7]	HHD	NA	Isometric	2	Average	NR	X	X					
Boeck et al (2012)[8b]	HHD	NA	Isometric	NR	NR	NR	X	X	X	X			
Cha et al (2014)[9]	Isokinetic dynamometer	60°/s[a] 180°/s	NR	NR	NR	NR		X	X				
Chaconas et al (2017)[10]	HHD	NA	Isometric	3	Peak	Body mass	X	X	X				
Dejaco et al (2016)[13]	HHD	NA	Isometric	3	Average	NR	X						
Dilek et al (2016)[16]	Isokinetic dynamometer	NA	Isometric	3	Average	NR	X	X					
Dubé et al (2023)[17]	HHD	NA	Isometric	3	Peak	Arm length	X	X					
Ellegaard et al (2016)[19b]	HHD	NA	Isometric	4	Peak	NR	X	X	X				
Elnour et al (2023)[20]	HHD	NA	Isometric	NR	NR	NR			X				
Ersever et al (2022)[21]	Isokinetic dynamometer	60°/s[a] 180°/s	Concentric	5-10	Peak	NR		X	X				
de Freitas et al (2014)[12]	HHD	NA	Isometric	3	Average	Body mass		X	X		X		
Hotta et al (2020)[25]	HHD	NA	Isometric	NR	NR	Body mass	X	X	X				
Ingwersen et al (2017)[26]	HHD	NA	Isometric	3	Average	Body mass		X	X		X		
Kanthanathan et al (2023)[27]	HHD	NA	Isometric	2	NR	NR	X	X					
Kara et al (2024)[28]	Isokinetic dynamometer	60°/s[a] 180°/s	Concentric	6	Peak	Body mass		X	X				
Karaaslan et al (2023)[29]	HHD	NA	Isometric	3	Average	NR	X	X	X	X			
Karabay et al (2025)[30]	HHD	NA	Isometric	3	Average	NR		X			X		
Lombardi et al (2008)[34]	Isokinetic dynamometer	60°/s[a] 180°/s	Concentric	NR	Peak	NR	X	X	X	X		X	X
Maenhout et al (2013)[36]	HHD	NA	Isometric	3	Average	NR		X	X		X		

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TABLE 3 Strength Testing Characteristics (continued)

Author (Year)	Strength Assessment Tool	Movement Speed	Contraction Type	Number of Trials	Outcome	Normalization Parameter	Movement Direction						
							ABD	ER	IR	FLX	SCAP	EXT	ADD
McClure et al (2004)[39]	HHD	NA	Isometric	NR	NR	NR		X	X		X		
Nyawose & Naidoo (2021)[45b]	HHD	NA	Isometric	NR	NR	NR	X	X	X	X			
Ortega-Cebrián et al (2021)[46]	HHD	NA	Isometric	NR	NR	NR		X	X				
Østerås & Torstensen (2010)[48b]	HHD	NA	Isometric	NR	NR	NR	X	X	X	X			
Rizzo et al (2017)[54]	Digital scale	NR	NR	4	Average	NR		X	X				
Schedler et al (2020)[57b]	HHD	NA	Isometric	2	Peak	NR		X	X				
Struyf et al (2013)[64]	HHD	NA	Isometric	NR	NR	NR					X		
Wang & Trudelle-Jackson (2006)[71]	HHD	NA	Isometric	2	Average	NR	X	X	X				

Abbreviations: ABD, abduction; ADD, adduction; ER, external rotation; EXT, extension; FLX, flexion; HDD, handheld dynamometer; IR, internal rotation; NA, not applicable; NR, not reported; SCP, scaption.
[a]*Included in the quantitative analysis.*
[b]*Excluded from the quantitative analysis.*

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excluding 20 study arms (36%) and 505 patients (35%) from the primary meta-analysis. This analysis reduced the strength of the effects and the heterogeneity of certain findings within specific intervention subgroups (**SUPPLEMENTAL TABLE 5** and **SUPPLEMENTAL FIGURES 8-12**). The overall effects of the meta-analyses and the subgroup effects for strengthening interventions were confirmed across all strength test directions, except for flexion. For multimodal interventions, effects were confirmed in all directions, except for scaption. The effects of education, sham/placebo, and natural course were confirmed in all strength test directions.

Secondary Analyses

The secondary analyses focused on strength test directions for abduction, external rotation, and internal rotation.

Strengthening Type **TABLE 5** and **SUPPLEMENTAL FIGURES 13-15** present the results of the secondary meta-analyses for strengthening type. The postintervention effects on abduction strength were large for NMES (1 arm; SMD, 0.85; 95% CI: 0.27, 1.43), moderate for concentric (2 arms; SMD, 0.65; 95% CI: 0.33, 0.96; $I^2 = 0\%$; $P = .99$) and eccentric (4 arms; SMD, 0.76; 95% CI: 0.05, 1.46; $I^2 = 85\%$; $P < .01$) training, and small for motor control (4 arms; SMD, 0.45; 95% CI: 0.18, 0.72; $I^2 = 0\%$; $P = .44$) and nonspecific (7 arms; SMD, 0.45; 95% CI: 0.24, 0.66; $I^2 = 0\%$; $P = .68$).

For external rotation strength, NMES (2 arms; SMD, 0.92; 95% CI: 0.48, 1.35; $I^2 = 0\%$; $P = .76$), eccentric (5 arms; SMD, 1.24; 95% CI: 0.25, 2.22; $I^2 = 93\%$; $P < .01$), and unaffected area (1 arm; SMD, 1.13; 95% CI: 0.46, 1.80) had large effects. Concentric (3 arms; SMD, 0.65; 95% CI: 0.14, 1.15; $I^2 = 59\%$; $P = .08$) has a moderate effect, and motor control (6 arms; SMD, 0.45; 95% CI: 0.22, 0.48; $I^2 = 0\%$; $P = .55$), nonspecific (9 arms; SMD, 0.38; 95% CI: 0.19, 0.56; $I^2 = 0\%$; $P = .83$), and low load (1 arm; SMD, 0.40; 95% CI: 0.01, 0.79) had small effects. High load (1 arm; SMD, 0.33; 95% CI: −0.07, 0.73) and blood flow restriction (1 arm; SMD, 0.67; 95% CI: −0.09, 1.43) had no effect on external rotation strength.

For internal rotation strength, NMES (2 arms; SMD, 0.95; 95% CI: 0.5, 1.39; $I^2 = 0\%$; $P = .84$) and unaffected area (1 arm; SMD, 1.24; 95% CI: 0.56, 1.92) had large effects; high load (1 arm; SMD, 0.50; 95% CI: 0.10, 0.90) had a moderate effect; and motor control (4 arms; SMD, 0.41; 95% CI: 0.10, 0.73; $I^2 = 0\%$; $P = .76$), nonspecific (9 arms; SMD, 0.35; 95% CI: 0.15, 0.55; $I^2 = 0\%$; $P = .98$), and low load (1 arm; SMD, 0.47; 95% CI: 0.07, 0.86) had small effects. Eccentric (2 arms; SMD, 0.32; 95% CI: −0.05, 0.70; $I^2 = 0\%$; $P = .63$) and blood flow restriction (1 arm; SMD, 0.54; 95% CI: −0.21, 1.30) had no effect on internal rotation strength.

Active Intervention Length **TABLE 6** and **SUPPLEMENTAL FIGURES 16-18** present the results of the secondary meta-analyses for active intervention length. The

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TABLE 4 Primary Meta-analysis

Intervention Category	Movement Direction																			
	Abduction			Adduction			External Rotation			Extension			Flexion			Internal Rotation			Scaption	
	n[a]	I^2	SMD (95% CI)	n[a]	I^2	SMD (95% CI)	n[a]	I^2	SMD (95% CI)	n[a]	I^2	SMD (95% CI)	n[a]	I^2	SMD (95% CI)	n[a]	I^2	SMD (95% CI)	n[a]	I^2	SMD (95% CI)
Strengthening	18	53%[b]	**0.58 (0.40, 0.77)**	1	…	0.26 (−0.25, 0.76)	29	76%[b]	**0.64 (0.43, 0.86)**	2	0%	0.16 (−0.27, 0.58)	4	0%	**0.38 (0.08, 0.67)**	21	0%	**0.47 (0.34, 0.59)**	6	52%	**0.58 (0.26, 0.91)**
Multimodal	4	0%	**0.29 (0.03, 0.55)**	…	…	…	16	80%[b]	**0.57 (0.23, 0.92)**	1	…	0.07 (−0.73, 0.87)	1	…	0.03 (−0.77, 0.83)	13	48%[b]	**0.46 (0.22, 0.69)**	3	0%	**0.31 (0.03, 0.60)**
Education	1	…	0.22 (−0.21, 0.66)	…	…	…	1	…	0.09 (−0.35, 0.52)	…	…	…	…	…	…	…	…	…	…	…	…
Sham/placebo	1	…	0.14 (−0.22, 0.49)	…	…	…	1	…	0.04 (−0.32, 0.39)	…	…	…	…	…	…	1	…	−0.02 (−0.38, 0.33)	…	…	…
Natural course	1	…	−0.07 (−0.58, 0.44)	1	…	0.03 (−0.47, 0.54)	2	0%	−0.01 (−0.43, 0.41)	1	…	…	1	…	−0.03 (−0.53, 0.48)	2	0%	0.01 (−0.41, 0.43)	…	…	…
OVERALL	25	54%[b]	**0.47 (0.32, 0.63)**	2	0%	0.14 (−0.21, 0.50)	49	78%[b]	**0.56 (0.39, 0.74)**	4	0%	0.05 (−0.25, 0.36)	6	0%	**0.25 (0.01, 0.49)**	37	25%	**0.41 (0.30, 0.52)**	9	35%	**0.48 (0.27, 0.70)**

Intervention arms were subgrouped based on the content of the intervention. Bold fonts indicate statistically significant confidence intervals.
Abbreviations: CI, confidence interval; SMD, standardized mean difference.
[a]*Number of arms.*
[b]*Heterogeneity analysis, P<.05.*

postintervention effect on abduction strength was moderate for intervention lengths of 8 (9 arms; SMD, 0.50; 95% CI: 0.31, 0.69; $I^2 = 0\%$; $P = .80$) and 12 (8 arms; SMD, 0.53; 95% CI: 0.19, 0.88; $I^2 = 80\%$; $P < .01$) weeks and small for 6 weeks (6 arms; SMD, 0.46; 95% CI: 0.18, 0.74; $I^2 = 14\%$; $P = .32$).

For external rotation strength, moderate effects were observed with 6 (11 arms; SMD, 0.55; 95% CI: 0.21, 0.88; $I^2 = 63\%$; $P < .01$), 9 (2 arms; SMD, 0.69; 95% CI: 0.30, 1.07; $I^2 = 0\%$; $P = .66$), and 12 (17 arms; SMD, 0.74; 95% CI: 0.37, 1.11; $I^2 = 90\%$; $P < .01$) weeks of intervention duration, and a small effect was observed with 8 weeks of intervention duration (16 arms; SMD, 0.47; 95% CI: 0.31, 0.62; $I^2 = 0\%$; $P = .66$).

For internal rotation strength, a moderate effect was observed with a 9-week intervention (2 arms; SMD, 0.51; 95% CI: 0.13, 0.89; $I^2 = 0\%$; $P = .77$), and small effects were observed with 6 (11 arms; SMD, 0.48; 95% CI: 0.25, 0.71; $I^2 = 29\%$; $P = .17$), 8 (14 arms; SMD, 0.43; 95% CI: 0.26, 0.60; $I^2 = 5\%$; $P = .40$), and 12 (8 arms; SMD, 0.41; 95% CI: 0.17, 0.64; $I^2 = 56\%$; $P = .02$) weeks of intervention.

DISCUSSION

Our findings complement previous systematic reviews that focused on the effect of rehabilitation interventions on pain and function recovery.[32,43] We included 56 intervention arms from 28 studies in the quantitative synthesis, providing a comprehensive coverage of the primary outcome effect. Overall, the primary meta-analyses showed improvements in shoulder external rotation, internal rotation, abduction, flexion, and scaption strength following rehabilitation interventions. Interventions that included active exercises (strengthening and multimodal) led to greater within-group improvements in shoulder strength compared to interventions without exercises (education, sham/placebo, natural course). Interventions focused on

TABLE 5 Secondary Meta-analyses

	Movement Direction								
	Abduction			External Rotation			Internal Rotation		
Strengthening Subgroup	n[a]	I^2	SMD (95% CI)	n[a]	I^2	SMD (95% CI)	n[a]	I^2	SMD (95% CI)
Motor control	4	0%	**0.45 (0.18, 0 .72)**	6	0%	**0.45 (0.22, 0.48)**	4	0%	**0.41 (0.10, 0.73)**
Nonspecific	7	0%	**0.45 (0.24, 0.66)**	9	0%	**0.38 (0.19, 0.56)**	9	0%	**0.35 (0.15, 0.55)**
Concentric	2	0%	**0.65 (0.33, 0.96)**	3	59%	**0.65 (0.14, 1.15)**	...	...	...
Eccentric	4	85%[b]	**0.76 (0.05, 1.46)**	5	93%[b]	**1.24 (0.25, 2.22)**	2	0%	0.32 (−0.05, 0.70)
NMES	1	...	**0.85 (0.27, 1.43)**	2	0%	**0.92 (0.48, 1.35)**	2	0%	**0.95 (0.51, 1.39)**
High load	...	...	...	1	...	0.33 (−0.07, 0.73)	1	...	**0.50 (0.10, 0.90)**
Low load	...	...	...	1	...	**0.40 (0.01, 0.79)**	1	...	**0.47 (0.07, 0.86)**
BFR	...	...	...	1	...	0.67 (−0.09, 1.43)	1	...	0.54 (−0.21, 1.30)
Unaffected area	...	...	...	1	...	**1.13 (0.46, 1.80)**	1	...	**1.24 (0.56, 1.92)**
OVERALL	18	53%[b]	**0.58 (0.40, 0.77)**	29	76%[b]	**0.64 (0.43, 0.86)**	21	0%	**0.47 (0.34, 0.59)**

Intervention arms classified as strengthening were subgrouped based on the type of strengthening exercises performed. Bold fonts indicate statistically significant confidence intervals.
Abbreviations: BFR, blood flow restriction; CI, confidence interval; NMES, neuromuscular electrical stimulation; SMD, standardized mean difference.
[a]*Number of arms.*
[b]*Heterogeneity analysis, P<.05.*

TABLE 6 Secondary Meta-analyses

	Movement Direction								
	Abduction			External Rotation			Internal Rotation		
Intervention Length	n[a]	I^2	SMD (95% CI)	n[a]	I^2	SMD (95% CI)	n[a]	I^2	SMD (95% CI)
6 weeks	6	14%	**0.46 (0.18, 0.74)**	11	63%[b]	**0.55 (0.21, 0.88)**	11	29%	**0.48 (0.25, 0.71)**
8 weeks	9	0%	**0.50 (0.31, 0.69)**	16	0%	**0.47 (0.31, 0.62)**	14	5%	**0.43 (0.26, 0.60)**
9 weeks	...	...	...	2	0%	**0.69 (0.30, 1.07)**	2	0%	**0.51 (0.13, 0.89)**
12 weeks	8	80%[b]	**0.53 (0.19, 0.88)**	17	90%[b]	**0.74 (0.37, 1.11)**	8	56%[b]	**0.41 (0.17, 0.64)**
OVERALL	23	52%[b]	**0.51 (0.35, 0.67)**	46	78%[b]	**0.60 (0.42, 0.78)**	35	24%	**0.43 (0.32, 0.55)**

All active intervention arms were subgrouped based on the length of the active intervention. Bold fonts indicate statistically significant confidence intervals.
Abbreviations: CI, confidence interval; SMD, standardized mean difference.
[a]*Number of arms.*
[b]*Heterogeneity analysis, P<.05.*

strengthening showed the largest within-group effects on abduction, external rotation, flexion, internal rotation, and scaption strength. The primary meta-analysis results were robust for abduction, external rotation, and internal rotation strength test directions, but less stable for flexion and scaption.

Several factors may affect strength recovery in people with RCRSP. People with high irritability may be unable to tolerate the loading required for muscle strengthening.[40] Shoulder strength gains might be also related to patients' presentation at the initial evaluation (eg, individuals with greater shoulder weakness might have more potential for improvement than those without such weakness). Strength gains depend on intervention-specific factors, including dosage, frequency, and load.[41] There is variability in exercise prescription for patients with shoulder pain.[18] Strength may not improve in some individuals with RCRSP due to patient-specific factors, underloaded exercises,[52] or insufficient frequency.[22] The goal of resistance exercises in rehabilitation is not always to increase strength. Resistance exercise parameters can be tailored to target other objectives, such as enhancing tendon load and promoting tissue healing.[37] Although beyond the scope of this systematic review,

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improvements in strength do not always translate to improvements in pain and function, and vice versa. This aligns with the biopsychosocial model of pain, which recognizes that, in some cases, symptom improvements may occur independently of mechanical or physiological changes.[52]

Some strengthening approaches, such as NMES, may produce larger within-group effects than others, although these findings are supported by a limited number of study arms, and should be interpreted cautiously. For example, NMES had large within-group effects on strength across all movement directions. However, these findings are based on 2 study arms. Although NMES was the most extensively studied modality for addressing muscle inhibition, particularly in the lower limbs,[44] its effects on strength in individuals with RCRSP remain understudied.[44] Eccentric exercises showed the greatest within-group improvement in external rotation strength, suggesting that incorporating eccentric exercises into rehabilitation programs may be particularly beneficial for addressing external rotator muscle weakness.

Intervention programs lasting between 6 and 12 weeks were effective at improving shoulder strength in individuals with RCRSP. Strengthening occurs through neural and hypertrophy mechanisms.[68] Pain inhibits these mechanisms through muscle inhibition and disuse,[67,70] blunting force output.[61,62] Individuals with shoulder pain have lower activation of the rotator cuff muscles,[59] and a strong relationship exists between voluntary activation and infraspinatus force.[61] Muscle strength can improve quickly after pain relief, suggesting that early changes in muscle strength are likely due to improvement in the neural mechanisms of muscle contractions.[59] Structural changes in shoulder muscles can also occur following exercises. In healthy individuals, supraspinatus[55] and infraspinatus[38,55] cross-sectional areas significantly increase following 8 weeks of exercise. Although studies included in our review measured strength output, none assessed muscle activation or structural changes. This limits understanding on the primary mechanisms underlying the strength recovery of patients with RCRSP, which should be investigated in future studies.

Moderate-to-high heterogeneity for abduction and external rotation strength in the primary analysis may stem from multiple sources, including strength assessments (tools and parameters) and intervention characteristics (frequency, intensity, type, and duration). Although most studies used a handheld dynamometer to assess strength, some studies employed isokinetic dynamometers. Additionally, different types of strength (isometric, isokinetic) were tested, and the number of repetitions per strength test varied considerably. Whereas most studies conducted 2 or 3 trials for a single strength direction, some reported 4-10 maximal trials. Outcome metrics also differed, with studies reporting either average or peak values, and some normalized strength using different

TABLE 7 Postintervention Strength Changes Expressed in Strength Units of Measurement

	Force				Torque				Normalized							
	kg[a]				N·m				kg/kg				N/kg			
Strength Direction	N[b]	SDp	ST	MT	N[b]	SDp	ST	MT	N[b]	SDp	ST	MT	N[b]	SDp	ST	MT
Abduction	4	4.35	1.74-3.35	0.13-2.39	4	22.42	8.97-17.27	0.67-12.33	2	0.05	0.02-0.04	0-0.03	...	...	...	...
External rotation	8	3.24	1.39-2.79	0.75-2.98	6	18.98	8.16-16.32	4.36-17.46	3	0.05	0.02-0.05	0.01-0.05	2	0.09	0.04-0.08	0.02-0.09
Flexion	1	2.04	0.16-1.37	−1.57 to 1.69[c]	1	11.81	0.94-7.91	−9.09 to 9.80[c]	...	...	...	...	...	...	...	...
Internal rotation	8	3.87	1.32-2.29	0.85-2.67	4	9.45	3.21-5.57	2.08-6.52	3	0.09	0.03-0.05	0.02-0.06	2	0.15	0.05-0.09	0.03-0.10
Scaption	3	4.93	1.28-4.49	0.15-2.96	...	...	...	...	1	0.09	0.02-0.08	0-0.05	2	0.21	0.05-0.19	0.01-0.12
Extension	1	1.34	−0.36 to 0.77[c]	−0.98 to 1.16[c]	1	18.17	−4.91 to 10.54[c]	−13.27 to 15.81[c]	...	...	...	...	...	...	...	...
Adduction	...	...	...	...	1	17.18	−4.27 to 13.06[c]	...	...	...	...	...	...	...	...	...

The range is calculated using the lower and upper bounds of the 95% confidence interval from the primary meta-analyses.
Abbreviations: kg, kilogram; N·m, newton meter; MT, multimodal intervention; SDp, pooled standard deviation; ST, strengthening intervention.
[a]*Studies that reported force values measured in newtons or pounds were converted into kilograms.*
[b]*Number of studies that measured strength in the specific direction and with the unit of measurement. These data were used to calculate the associated pooled standard deviation.*
[c]*Effect of the intervention on the strength direction is nonsignificant.*

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parameters (eg, body mass or arm length). Intervention variability may also further contribute to heterogeneity. There were differences in exercise program characteristics, including frequency, intensity, type, and duration, which is similar to a recent scoping review on intervention characteristics for patients with RCRSP.[18]

Among the randomized controlled trials, the highest risk of bias was observed in the domains of outcome measurement and randomization. In the outcome domain, the main sources of bias were the lack of assessor blinding and the absence of standardized measurements reporting. In the randomization domain, the risk of bias was primarily due to unreported allocation concealment. These factors may reduce confidence in the study findings. For prospective longitudinal studies, the methodological quality was classified as poor, mainly due to the absence of a control group.

Clinical Implications

Active rehabilitation interventions had a positive effect on shoulder strength. However, our meta-analyses did not directly compare different interventions, and therefore, our results cannot support the superiority of 1 intervention over another when applied under similar conditions. **TABLE 7** reports shoulder strength changes expressed in strength units of measurement, providing useful benchmarks to assess strength recovery in clinical settings. The secondary analysis showed that clinicians can tailor strengthening intervention content or length to patient preferences.

Strengths and Limitations

The use of within-arm statistical analysis is a strength of this review, as it enabled the inclusion of both randomized controlled trials and cohort studies, offering broader coverage of treatment effects. However, this approach also has limitations. Including nonrandomized studies lowers the overall certainty of evidence, and within-group analyses do not allow for direct comparison between intervention categories or types of strengthening exercise. Additionally, findings related to interventions that do not include exercises may be limited by the small number of study arms and the fact that only control interventions from clinical trials were included. The small number of study arms also affects the secondary meta-analyses. Another limitation is the statistical heterogeneity in some strength measures and subgroups in both primary and secondary meta-analyses. Last, our systematic review may underestimate the number of patients with rotator cuff tears included in the sample. Thirteen studies did not report whether they included or excluded patients with rotator cuff tears.

Future Directions

Future studies of RCRSP should clearly specify whether the study sample includes individuals with rotator cuff tears. Additional high-quality studies with larger sample sizes are needed to strengthen the evidence surrounding strength recovery in patients with RCRSP, particularly for shoulder flexion, scaption, and abduction movement directions.

CONCLUSION

Interventions incorporating active exercises had the largest effect on shoulder strength in individuals with RCRSP. Strengthening interventions that include eccentric exercises and NMES may yield the most substantial improvements in shoulder abduction, external rotation, and internal rotation strength. Additionally, within-group changes in these strength measures appear consistent across interventions of at least 6-12 weeks in length. ◉

KEY POINTS

FINDINGS: Interventions with active exercises had moderate effects on shoulder strength in individuals with rotator cuff–related shoulder pain (RCRSP). Comparator interventions that did not include active exercises—most of which served as control conditions in the clinical trials reviewed—did not result in significant strength gains in individuals with RCRSP.

IMPLICATIONS: Strengthening interventions that include eccentric exercises and neuromuscular electrical stimulation may yield the most substantial improvements in shoulder abduction, external rotation, and internal rotation strength. Strengthening interventions between 6 and 12 weeks in length had similar beneficial effects on abduction, external rotation, and internal rotation strength. Intervention length may not be a critical factor when prescribing exercise for patients with RCRSP.

CAUTION: A within-group design does not allow for a direct comparison between intervention categories, strengthening exercise types, and intervention length.

STUDY DETAILS

AUTHOR CONTRIBUTIONS: B.Z. and I.A.R. equally contributed to the article and are considered co-first authors. B.Z., I.A.R., and F.P. were involved in the conception of the study. B.Z., I.A.R., S.G., J.M.-D., J.-S.R., and F.P. were involved in the design of the study. B.Z., I.A.R., J.M.-D., and F.P. were involved in data acquisition. B.Z., I.A.R., F.D., and F.P. were involved in data extraction. B.Z., I.A.R., S.G., and F.P. were involved in data analysis. B.Z., I.A.R., S.G., J.-S.R., and F.P. were involved in data interpretation. B.Z., I.A.R., J.M.-D., S.G., and F.P. were involved in manuscript drafting. B.Z., I.A.R., F.D., J.M.-D., S.G., J.-S.R., and F.P. were involved in manuscript reviewing and editing. All authors approved the final version of the article. F.P. was responsible for obtaining project funding and takes responsibility for the integrity of the work as a whole. All authors have read and agreed to the published version of the article.

DATA SHARING: Data are available upon request. Data extraction spreadsheets, data used for the analysis, and data analysis codes are available upon

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reasonable request by contacting the corresponding author. Data may be reused for research aims.

PATIENT AND PUBLIC INVOLVEMENT: Patients and the public were not involved in the research.

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